

## FlexZone — Flowcharts (Text / Defense-Ready)

Convention: **START** / **END**; **Decision** = diamond in drawn figure; **Process** = rectangle.

These flows reflect **implemented** behavior at a high level.

### 1. Member Registration (Join + Portal Account)

START

- Visitor opens Join wizard
- Process: Enter branch, plan context, personal details
- Process: Complete Turnstile CAPTCHA
- Decision: Validation & CAPTCHA OK?
  - NO → Show errors → END (retry)
  - YES → Process: Create member record (pending / unverified)
- Process: Show success / redirect to login or instructions

END

*Registered members who complete email verification (if enabled) may then use member login.*

### 2. Online Payment (Xendit)

START

- Member logged in
- Process: Select plan, duration, payment method = online
- Decision: Active subscription blocks new purchase?
  - YES → Show message → END
  - NO → Continue
- Process: Create pending payment; create Xendit invoice
- Member pays at Xendit
- Xendit → Webhook POST
- Decision: Token valid & status PAID/SETTLED?
  - NO → Log / ignore → END (webhook still 200 if required)
  - YES → Process: Finalize payment (paid), create subscription, sales row, card/QR, notifications, email,

audit

END

### 3. Login & Lockout (All Roles)

START

- User enters credentials

→ Process: Complete Turnstile CAPTCHA  
→ Decision: CAPTCHA valid?  
    NO → Increment Turnstile lockout → END  
    YES → Continue  
→ Decision: Account Locked? (Check login\_attempts)  
    YES → Show lockout message (time remaining) → END  
    NO → Continue  
→ Decision: Credentials correct?  
    NO → Increment Login Lockout; Log failure → END  
    YES → Process: Reset attempt counter; Create session; Log success  
END

#### **4. Account Expiration & Archive Purge**

START (Scheduled / Triggered)  
→ Process: Query unverified members > 7 days old  
→ Decision: Found?  
    YES → Process: Move to archived\_members (reason: Expired); Log action  
    NO → Continue  
→ Process: Query archived members > 30 days residency  
→ Decision: Found?  
    YES → Process: Permanently delete records; Log action  
    NO → END  
END

#### **5. Cash Payment (Pay at Branch)**

START  
→ Member selects cash/manual  
→ Process: Create pending payment (manual method)  
→ Member pays at branch (off-system money handling)  
→ Staff opens admin action: confirm cash  
→ Decision: Pending manual payment exists & no conflicting active sub?  
    NO → Error message → END  
    YES → Process: Finalize payment (paid, staff-confirmed), subscription, sales, card/QR, notifications, audit  
END

## 6. Subscription Activation (Shared Pipeline)

START

- Input: payment id marked paid (online or cash path)
- Process: Compute dates from plan duration on payment
- Process: Insert subscription (active)
- Process: Link payment ↔ subscription; sync member cache
- Process: Ensure membership card + QR payload
- Process: Notification (payment + subscription messages per code)
- Decision: Email send configured?
  - SKIP detail in flowchart if always “attempt send”

END

## 7. Monitoring Scan

START

- Input: Scan string (QR token / member id)
- Process: Resolve member record
- Decision: Found?
  - NO → Deny → END
  - YES → Process: Validate token ownership (if applicable)
- Decision: Subscription active & end date valid?
  - NO → Deny / show expired → END
  - YES → Process: Determine IN vs OUT
- Decision: Duplicate scan window?
  - YES → Ignore / message → END
  - NO → Process: Insert monitoring\_logs
- Show success

END

## 8. Appointment Submission (Public)

START

- Visitor opens appointment form
- Process: Fill branch, type, date, time, contact fields
- Decision: Valid?
  - NO → Errors → END
  - YES → Process: Save appointment request
- Confirmation message

END

## 9. Inquiry Submission (Public)

START

- Visitor opens contact form
- Process: Enter name, email, message
- Decision: Valid?
  - NO → Errors → END
  - YES → Process: Save inquiry
- Confirmation

END

## 10. Report Generation (Admin/Staff)

START

- User authenticated (role permits)
- User opens Dashboard / Reports
- Process: API queries DB (members, subs, sales, monitoring, ...)
- Process: Render charts / tables (JS)
- Decision: Export requested?
  - YES → Process: CSV/PDF per export API
  - NO → END

END

## 11. Notification Flow (Member)

START

- Trigger event (payment success, activation, expiry, promo, ID ready, ...)
- Process: insert row into notifications (valid type enum)
- Member opens portal
- Process: Load recent notifications; unread count
- Decision: Mark read?
  - YES → Update is\_read

END

## 12. Audit Log Flow (Staff/System)

START

- Staff action or system event (e.g. online payment finalized)
- Decision: Audited action type?
  - NO → END
  - YES → Process: INSERT audit\_logs (user if session; details JSON)

→ Admin reviews system logs / audit UI  
END

*For ISO 25010 evaluation, map each flow to suitability, reliability, and security evidence (test logs, screenshots, checklist).*